An explicit tenth-discriminant bound from Banks's zero-free theorem

Research note prepared with Codex

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Abstract

Taking the explicit large-conductor theorem of Banks (`arXiv:2610.15483v1`, Theorem 1.1) as an external input, we show that an imaginary quadratic field of class number one has absolute fundamental discriminant less than 10^{10}. The additional analytic argument is a short, explicit application of Poisson summation to the integer lattice of the field. An exhaustive integer calculation using only primes at most 109 then recovers exactly the nine familiar fundamental discriminants. Two implementations, using different algorithms, give the same result. The argument does not use Heegner's solution, Heegner points, or a precomputed class-number table.

1 Statement and dependency

Throughout, $D < 0$ is a *fundamental discriminant*, $q = |D|$, $K = \mathbb{Q}(\sqrt{D})$, and χ_D is the primitive quadratic character of conductor q. All logarithms are natural. In particular, q is the absolute field discriminant, not necessarily the squarefree integer appearing under a square root.

The only recent theorem used is the following assertion of Banks [1]:

$$q \geq 10^{10}, \quad L(\beta, \chi) = 0, \quad 0 < \beta < 1 \quad \Longrightarrow \quad (1-\beta)\log q \geq \frac{3}{1000} \tag{1}$$

for every primitive real character χ of conductor q. Only this large-conductor assertion is needed. Its extension to smaller conductors by a separate computation is not used.

Theorem 1. *Assuming the external theorem (1), if $h(K) = 1$, then*

$$q < 10^{10}.$$

Combining this bound with the finite verification described in Section 3 gives

$$q \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Every discriminant on this list does have class number one.

The qualifier in the statement records the precise external dependency: this note does not supply an independent proof of Banks's theorem. The contribution here is the explicit passage from that theorem to the class-number-one problem, together with a reproducible finite exclusion. No novelty or optimality claim is made for the underlying classical theta-function argument.

2 Class number one produces a nearby real zero

Lemma 2. *If $q \geq 10^{10}$ and $h(K) = 1$, then $L(s, \chi_D)$ has a real zero*

$$1 - \frac{10}{\sqrt{q}} < \beta < 1.$$

Proof. Put

$$R = \sqrt{q}, \qquad y = \frac{R}{2}, \qquad \Lambda = y^{-1/2}\mathcal{O}_K \subset \mathbb{C} \simeq \mathbb{R}^2.$$

The embedded integer lattice $\mathcal{O}_K$ has covolume y, so Λ has covolume one. Define

$$\Theta(t) = \sum_{\alpha \in \mathcal{O}_K} \exp\left(-\frac{\pi t|\alpha|^2}{y}\right) \qquad (t > 0).$$

For any planar lattice of covolume one, its Euclidean dual is a rotation by 90 degrees of the original lattice. For example, if the columns of A are an oriented basis and $\det A = 1$, then $A^{-T} = JAJ^{-1}$, where J is rotation by 90 degrees and $J^{-1}\mathbb{Z}^2 = \mathbb{Z}^2$. Consequently, Poisson summation for the Gaussian gives

$$\Theta(t) = t^{-1}\Theta(1/t). \tag{2}$$

Since $q > 4$, the units of $\mathcal{O}_K$ are exactly $\{\pm 1\}$. Class number one implies that every nonzero integral ideal has exactly two generators in $\mathcal{O}_K$. Thus, initially for $\Re s > 1$,

$$\sum_{\alpha \in \mathcal{O}_K \setminus \{0\}} |\alpha|^{-2s} = 2\zeta_K(s).$$

Taking the Mellin transform and splitting its integral at $t = 1$, then using (2) in the part below 1, gives

$$2\Gamma(s)\left(\frac{y}{\pi}\right)^s \zeta_K(s) = -\frac{1}{s(1-s)} + \int_1^\infty (\Theta(t) - 1)\left(t^{s-1} + t^{-s}\right) dt. \tag{3}$$

The integral on the right converges locally uniformly for all complex s, since $\Theta(t) - 1$ decays exponentially as $t \to \infty$. Hence (3) continues the initial identity meromorphically. For clarity, the rational term is

$$\int_0^1 (t^{s-2} - t^{s-1})\, dt = \frac{1}{s-1} - \frac{1}{s} = -\frac{1}{s(1-s)}.$$

Set

$$\varepsilon = \frac{10}{R}, \qquad s_0 = 1 - \varepsilon, \qquad A = \frac{y}{\pi} = \frac{R}{2\pi}.$$

Here $R \geq 10^5$, $0 < s_0 < 1$, and $A > 1$. The two lattice points $\alpha = 1, -1$ alone give

$$\Theta(t) - 1 \geq 2e^{-\pi t/y} \geq \frac{2}{e} \qquad (1 \leq t \leq A).$$

Dropping the nonnegative t^{-s_0} term and the part of the integral above A, the integral in (3) is therefore at least

$$\frac{2}{e}\int_1^A t^{s_0 - 1}\, dt = \frac{2}{e\, s_0}(A^{s_0} - 1).$$

It suffices to prove

$$\frac{2\varepsilon}{e}(A^{1-\varepsilon} - 1) > 1. \tag{4}$$

All estimates needed for (4) can be made rational. The function $(\log R)/R$ decreases for $R > e$, and $\log(10^5) < 12$. Hence

$$0 < \varepsilon \log A \le \frac{10 \log R}{R} < \frac{3}{2500}.$$

Using $e^{-u} \ge 1 - u$,

$$A^{1-\varepsilon} = A e^{-\varepsilon \log A} > \frac{R}{2\pi} \frac{2497}{2500}.$$

The elementary bounds $\pi < 22/7$ and $2 < e < 3$ now imply

$$\begin{aligned} \frac{2\varepsilon}{e}(A^{1-\varepsilon} - 1) &> \frac{10}{\pi e} \frac{2497}{2500} - \frac{20}{eR} \\ &> \frac{35}{33} \frac{2497}{2500} - \frac{1}{10000} \\ &= \frac{31777}{30000} > 1. \end{aligned}$$

For an entirely rational check of the logarithmic bound, use $e > 8/3$ and $8^{12} > 10^5\, 3^{12}$. This proves (4), and so $\zeta_K(s_0) > 0$.

As $s \to 1^-$, the rational term in (3) tends to $-\infty$, while its integral stays finite. Thus $\zeta_K(s) < 0$ sufficiently close to 1 from the left. Continuity gives a zero $\beta \in (s_0, 1)$. Finally $\zeta_K(s) = \zeta(s) L(s, \chi_D)$, and $\zeta(s) \neq 0$ for real $0 < s < 1$. One elementary proof of the latter uses the positive alternating series

$$\eta(s) = \sum_{n \ge 1} (-1)^{n-1} n^{-s} > 0, \qquad \zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}} < 0.$$

Therefore $L(\beta, \chi_D) = 0$, as claimed. □

Proof of the analytic bound in Theorem 1. If $q \ge 10^{10}$ and $h(K) = 1$, Lemma 2 gives

$$(1 - \beta) \log q < \frac{10 \log q}{\sqrt{q}}.$$

The function $(\log q)/\sqrt{q}$ decreases for $q > e^2$. Using $\log(10^{10}) < 24$, we obtain

$$(1 - \beta) \log q < \frac{10 \cdot 24}{10^5} = \frac{3}{1250} < \frac{3}{1000}.$$

This contradicts (1). □

3 A finite exclusion with small primes

Lemma 3. *If $h(K) = 1$, then every prime $p < q/4$ is inert in K.*

Proof. If p splits or ramifies, there is an integral ideal of norm p. It is principal when $h(K) = 1$, so it has a generator $\alpha \in \mathcal{O}_K$ with $|\alpha|^2 = p$. A rational integer cannot have prime norm in a quadratic field. Every nonrational algebraic integer in K can be written $(a + b\sqrt{-q})/2$, with integers a, b and $b \neq 0$; the allowed parities depend on D. Consequently,

$$p = |\alpha|^2 = \frac{a^2 + q b^2}{4} \ge \frac{q}{4},$$

a contradiction. □

For $q > 8$, Lemma 3 applies to $p = 2$. Thus 2 is inert, D is odd, and $D \equiv 5 \pmod 8$; equivalently, $q \equiv 3 \pmod 8$. The negative fundamental discriminants with $q \leq 8$ are $-3, -4, -7, -8$.

For every remaining candidate, therefore, it is enough to examine

$$11 \leq q < 10^{10}, \qquad q \equiv 3 \pmod 8.$$

For an odd prime $p < q/4$, exclude q whenever $-q$ is zero or a square modulo p. The search deliberately includes nonfundamental q, so no squarefreeness test can accidentally discard a valid candidate.

Exact computation

Two programs accompany this note:

- `verify_sieve.cpp` builds permitted residue classes using the Chinese remainder theorem, then checks candidates individually using tables of squares modulo primes.
- `verify_independent.py` obtains the initial residue classes by direct scanning and Euler's criterion. It then checks whole arithmetic progressions using periodic integer bitsets.

Both first impose the conditions for $2, 3, 5, 7, 11, 13, 17$. The resulting wheel has modulus

$$M = 8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 2042040$$

and 1440 permitted residue classes. For $437 \leq q < 10^{10}$, every prime at most 109 satisfies $4p < q$. The programs check the remaining primes from 19 through 109, and find no surviving q. For $11 \leq q < 437$, they apply only the prime conditions with $4p < q$, and find exactly

$$11, \quad 19, \quad 43, \quad 67, \quad 163.$$

The endpoint 10^{10} is excluded analytically, and is the exclusive upper endpoint in both programs. These calculations use exact integer arithmetic throughout.

Checking the surviving fields

The Python verifier independently enumerates all primitive reduced positive binary quadratic forms of each remaining discriminant. The unique reduced form in each case is as follows:

q	3	4	7	8	11	19	43	67	163
(a, b, c)	$(1, 1, 1)$	$(1, 0, 1)$	$(1, 1, 2)$	$(1, 0, 2)$	$(1, 1, 3)$	$(1, 1, 5)$	$(1, 1, 11)$	$(1, 1, 17)$	$(1, 1, 41)$

In the enumeration, $|b| \leq a \leq c$, $b^2 - 4ac = -q$, $\gcd(a, b, c) = 1$, and $b \geq 0$ on the boundaries $|b| = a$ or $a = c$. The standard bound $a \leq \sqrt{q/3}$ makes the enumeration finite. The classical correspondence with ideal classes verifies class number one for these nine fields.

4 Reproduction, scope, and audit

Run the following commands from the directory containing the sources:

```
g++ -O2 -std=c++17 -Wall -Wextra -pedantic \
    verify_sieve.cpp -o verify_sieve
./verify_sieve
python3 verify_independent.py
```

The programs require no number-theory library or external data. Their outputs are saved in `sieve_cpp_output.txt` and `sieve_python_output.txt`. The Python program also checks the exact rational inequalities in the analytic argument.

The source snapshot and its SHA-256 hash are recorded in `sources/manifest.json`. The recent external dependency is Banks's stated Theorem 1.1 for $q \geq 10^{10}$; this note does not certify its full nineteen-page proof. Classical inputs are Poisson summation for a Gaussian, basic quadratic ideal theory and the factorization $\zeta_K = \zeta L$, and reduction of positive binary quadratic forms. The analytic lemma is proved above rather than imported with an unspecified error constant.

The constant 10 in Lemma 2 is chosen for a short proof with a comfortable margin. It is unrelated to the illustrative additive constant $C_{\mathrm{LS}} = 10$ in the blog comments. Sharpening this lemma alone would not lower the large-conductor threshold in the quoted form of Banks's theorem. The finite exclusion already gives the exact final classification.

This is a candidate contribution to Tao's challenge [2], with a clear external theorem dependency and reproducible arithmetic. It is not a claim to have independently established the new zero-free theorem, and no Lean formalization has been produced.

AI disclosure. The route selection, explicit estimates, proof text, and both verification programs were produced by Codex in this conversation. The two computations are algorithmically independent, but were written by the same AI assistant; they are not an independent human or independent-model review. The user has not yet reviewed or endorsed the mathematical argument. Nothing has been submitted to the blog or published externally.

References

[1] W. D. Banks, *On the OpenAI proof that Landau–Siegel zeros do not exist*, 11 October 2026, arXiv:2610.15483v1, Theorem 1.1. https://arxiv.org/html/2610.15483v1.

[2] T. Tao, *An experimental AI-math challenge: using tenth discriminant bounds to explore arguments ruling out Siegel zeroes*, 20 October 2026. Online challenge.